

2. Advanced publish

Nextflow from zero to hero

2026/01/28

Output files in a pipeline

Work vs Publish

- By default, files created by a process stay in the working directory
- We can *publish* them to another location using the `publishDir` directive
 - rename
 - decide how to publish
 - control which files are published
 - organise in logical directory structures

publishDir

```
process CREATE_FILES {  
  publishDir 'data/'  
  
  output:  
  path 'sample*'  
  
  script:  
  """  
  SAMPLES=("sample1" "sample2")  
  EXTS=("csv" "fasta" "bed")  
  
  for s in "${SAMPLES[@]"; do  
    for e in "${EXTS[@]"; do  
      touch "${s}.${e}"  
    done  
  done  
  """  
}
```

CREATE_FILES

Work

sample1.csv, sample1.fasta, sample1.bed
sample2.csv, sample2.fasta, sample2.bed

Publish
(data/)

sample1.csv, sample1.fasta, sample1.bed
sample2.csv, sample2.fasta, sample2.bed

publishDir

```
process CREATE_FILES {  
  publishDir 'data/'  
  
  output:  
  path 'sample1*'  
  
  script:  
  """  
  SAMPLES=("sample1" "sample2")  
  EXTS=("csv" "fasta" "bed")  
  
  for s in "${SAMPLES[@]"; do  
    for e in "${EXTS[@]"; do  
      touch "${s}.${e}"  
    done  
  done  
  """  
}
```

CREATE_FILES

Work

sample1.csv, sample1.fasta, sample1.bed
sample2.csv, sample2.fasta, sample2.bed

Publish
(data/)

sample1.csv, sample1.fasta, sample1.bed
sample2.csv, sample2.fasta, sample2.bed

Only files that match the declaration in the output block are published, not all the outputs of the process.

publishDir

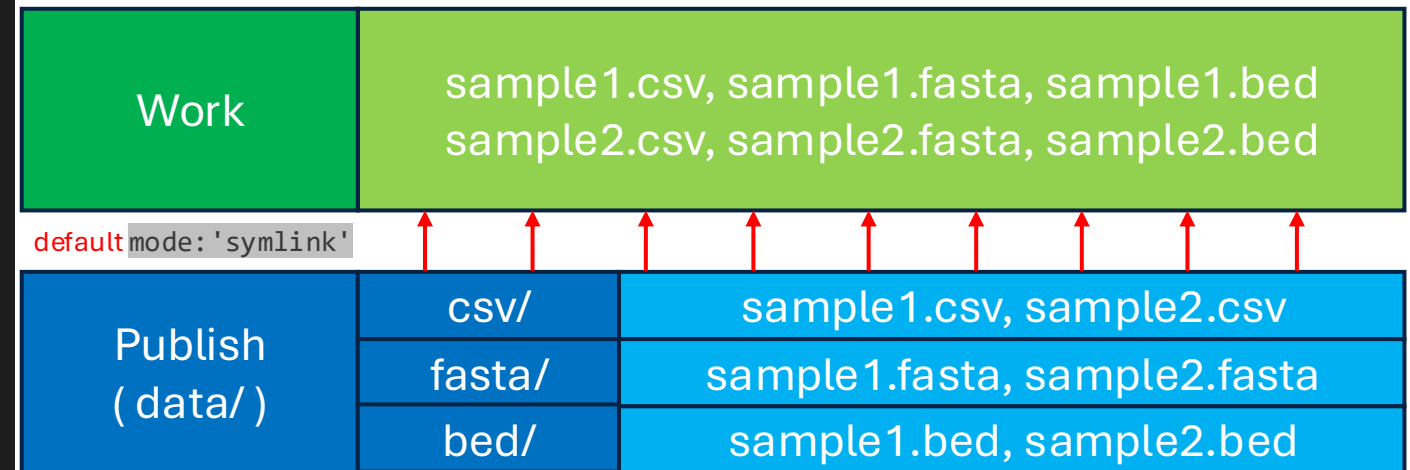
```
process CREATE_FILES {
  publishDir 'data/csv', pattern: "*.csv"
  publishDir 'data/fastq', pattern: "*.fastq"
  publishDir 'data/bed', pattern: "*.bed"

  output:
  path 'sample*'

  script:
  """
  SAMPLES=("sample1" "sample2")
  EXTS=("csv" "fastq" "bed")

  for s in "${SAMPLES[@]}"; do
    for e in "${EXTS[@]}"; do
      touch "${s}.${e}"
    done
  done
  """
}
```

CREATE_FILES



- `publishDir` can be specified more than once in order to publish output files to different target directories
- `pattern` selects which files to publish from the overall set of output files.

publishDir

```
process CREATE_FILES {
  publishDir 'data/csv',
    pattern: "*.csv", mode: "move"
  publishDir 'data/fastq',
    pattern: "*.fastq", mode: "copy"
  publishDir 'data/bed',
    pattern: "*.bed", mode: "copy"

  output:
  path 'sample*'

  script:
  """
  SAMPLES=("sample1" "sample2")
  EXTS=("csv" "fastq" "bed")

  for s in "${SAMPLES[@]}; do
    for e in "${EXTS[@]}; do
      touch "${s}.${e}"
    done
  done
  """
}
```

CREATE_FILES

Work	sample1.csv, sample1.fastq, sample1.bed sample2.csv, sample2.fastq, sample2.bed
------	--

Publish (data/)	csv/	sample1.csv, sample2.csv
	fastq/	sample1.fastq, sample2.fastq
	bed/	sample1.bed, sample2.bed

- Use `mode` to set the publishing method for each file

publishDir

```
process CREATE_FILES {
  publishDir 'data/csv',
    pattern: "*.csv", mode: "move",
    saveAs: { filename -> if (filename.contains('sample1')) return "sample_one.csv"
              else return "sample_two.csv"}
  publishDir 'data/fastq',
    pattern: "*.fastq", mode: "copy",
    saveAs: { filename -> if (filename.contains('sample1')) return "sample_uno.fastq"
              else return "sample_dos.fastq"}
  publishDir 'data/bed',
    pattern: "*.bed", mode: "copy",
    saveAs: { filename -> if (filename.contains('sample1')) return "sample_uno.bed"
              else return "sample_dos.bed"}

  output:
  path 'sample*'

  script:
  """
  SAMPLES=("sample1" "sample2")
  EXTS=("csv" "fastq" "bed")

  for s in "${SAMPLES[@]"; do
    for e in "${EXTS[@]"; do
      touch "\${s}.\${e}"
    done
  done
  """
}
```

CREATE_FILES

Work

sample1.csv, sample1.fasta, sample1.bed
sample2.csv, sample2.fasta, sample2.bed

Publish
(data/)

csv/

sample_one.csv, sample_two.csv

fasta/

sample_uno.fasta, sample_dos.fasta

bed/

sample_uno.bed, sample_dos.bed

- Use `saveAs` to set the name of the published files.

publishDir

...dinamicamente

```
process FASTP {
  publishDir "${params.outdir}/processed_reads/${sample_id}",
    pattern: '*-qced.fastq.gz',
    saveAs: { filename ->
      if (filename.contains('R1')) return "${sample_id}_R1.fastq.gz"
      else if (filename.contains('R2')) return "${sample_id}_R2.fastq.gz"
      else return filename
    }

  publishDir "${params.outdir}/qc_reports/${sample_id}",
    pattern: '*.json',
    saveAs: { _filename -> "${sample_id}_qc_report.json" }

  publishDir "${params.outdir}/qc_reports/${sample_id}",
    pattern: '*.html',
    saveAs: { _filename -> "${sample_id}_qc_report.html" }

  input:
  tuple val(sample_id), val(fastq_set_id), path(fastq_R1), path(fastq_R2)

  // ...
}
```